

Source: Johnny Briggs on Unsplash

HARNESS ENGINEERING IS NOT NEW

CARLY RICHMOND

-
-
-



Source: JD-Photos on Unsplash

ABOUT ME

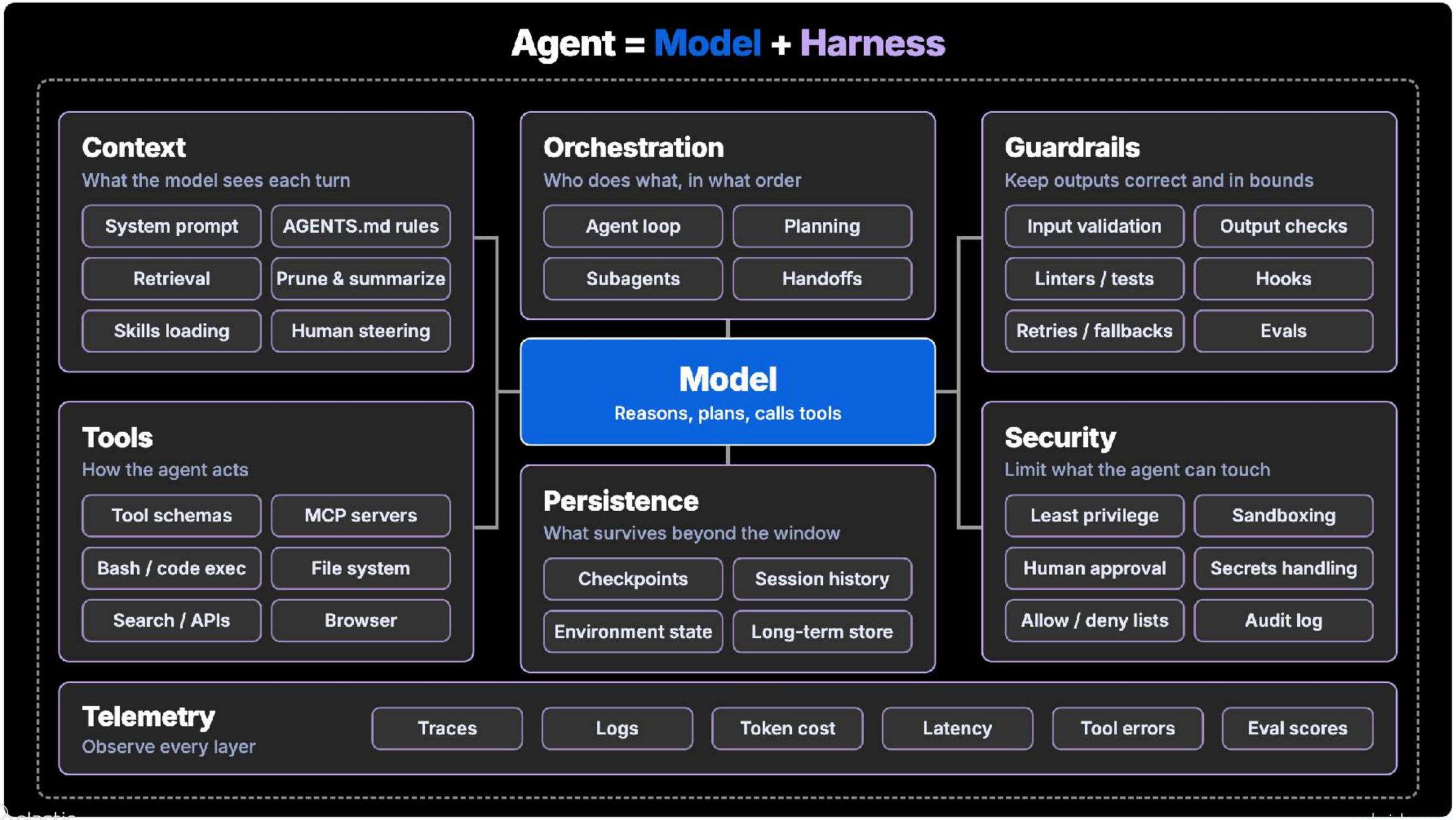
- Senior Manager, Developer Advocacy @ elastic
- Frontend Engineer, Speaker & Blogger

SCAN ME

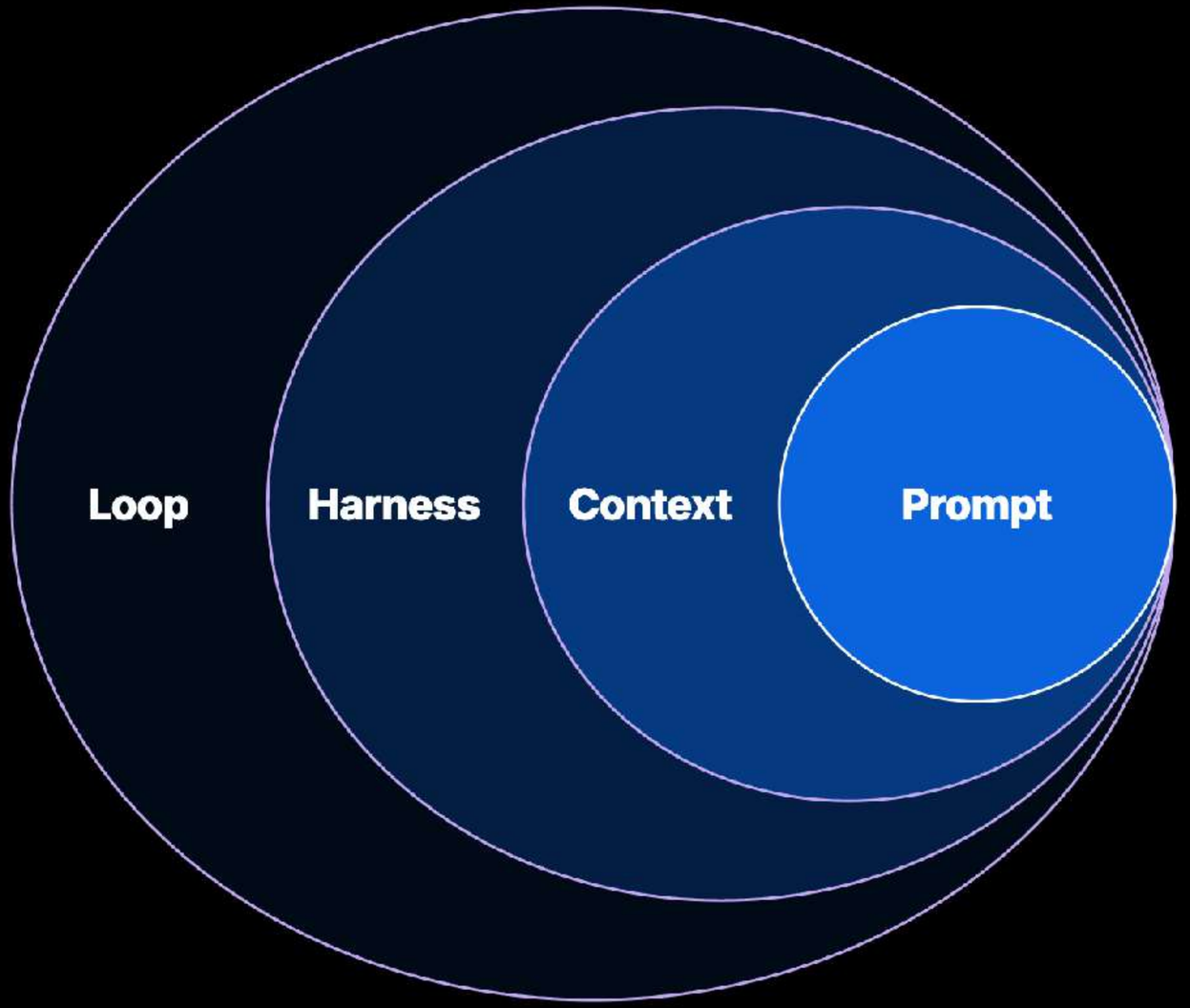


WHAT IS HARNESS ENGINEERING?




Agent = Model + Harness
Context
What the model sees each turn
System prompt
AGENTS.md rules
Retrieval
Prune & summarize
Skills loading
Human steering
Orchestration
Who does what, in what order
Agent loop
Planning
Subagents
Handoffs
Guardrails
Keep outputs correct and in bounds
Input validation
Output checks
Linters / tests
Hooks
Retries / fallbacks
Evals
Model
Reasons, plans, calls tools
Tools
How the agent acts
Tool schemas
MCP servers
Bash / code exec
File system
Search / APIs
Browser
Persistence
What survives beyond the window
Checkpoints
Session history
Environment state
Long-term store
Security
Limit what the agent can touch
Least privilege
Sandboxing
Human approval
Secrets handling
Allow / deny lists
Audit log
Telemetry
Observe every layer
Traces
Logs
Token cost
Latency
Tool errors
Eval scores

Loop, Harness, Context and Prompt Engineering are all related



1 Prompt engineering

Creating and optimising the instructions and constraints from users and agent developers.

2 Context engineering

Managing the LLM context window using instructions, memory, tools and structured outputs.

3 Harness engineering

Developing and maintaining the full operating environment, software infrastructure, and feedback loops around an AI model to build reliable, autonomous agents.

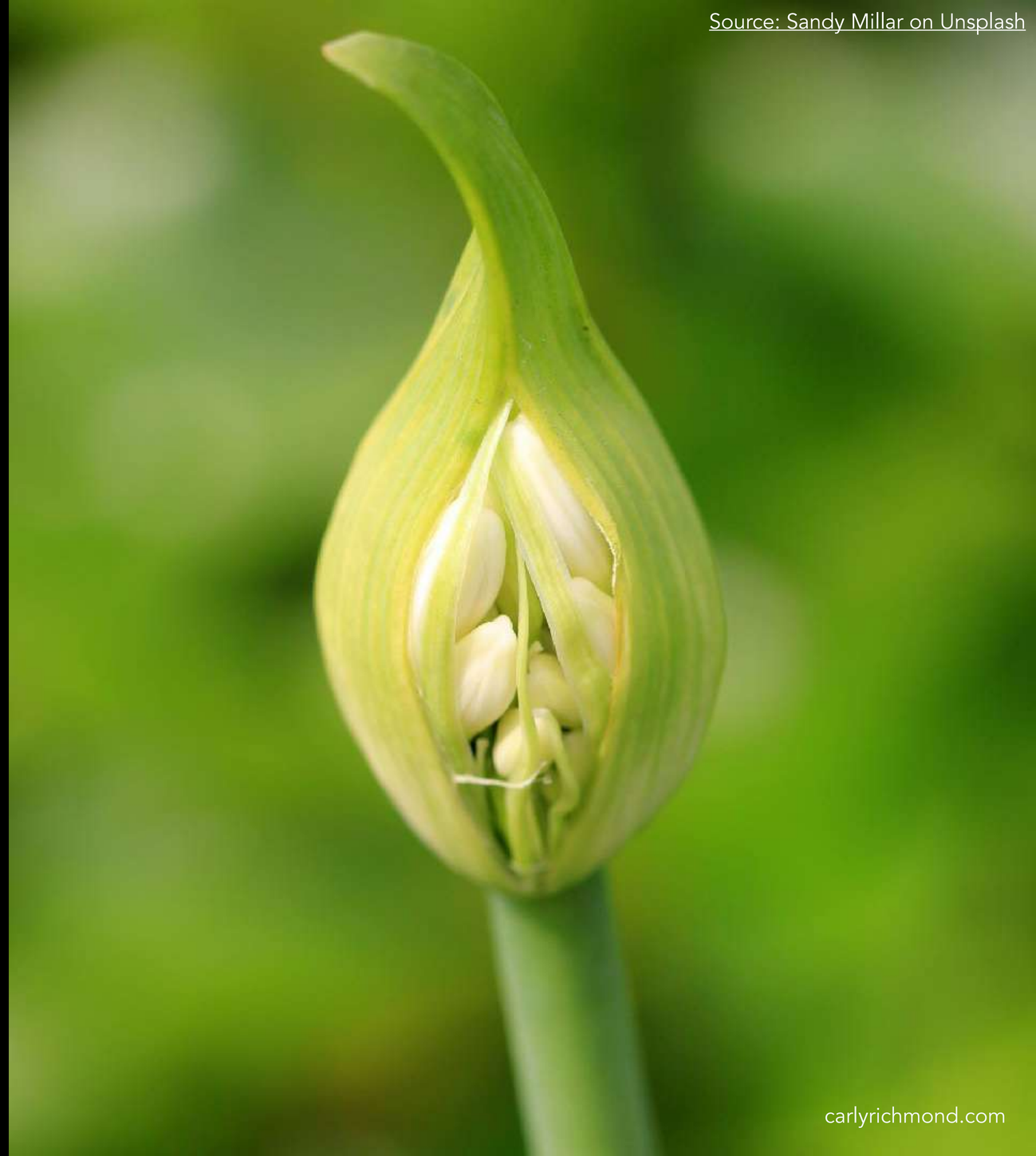
4 Loop engineering

Defining the goals, conditions and stop criteria that allow the agent to execute, adapt and adjust itself without intervention.

elastic
carlyrichmond.com

WHAT'S NEW?

- Non-determinism
- Prompting
- Reliance on text output
- Model versioning
- Context window and memory management
- AI-driven software development
- Token costs



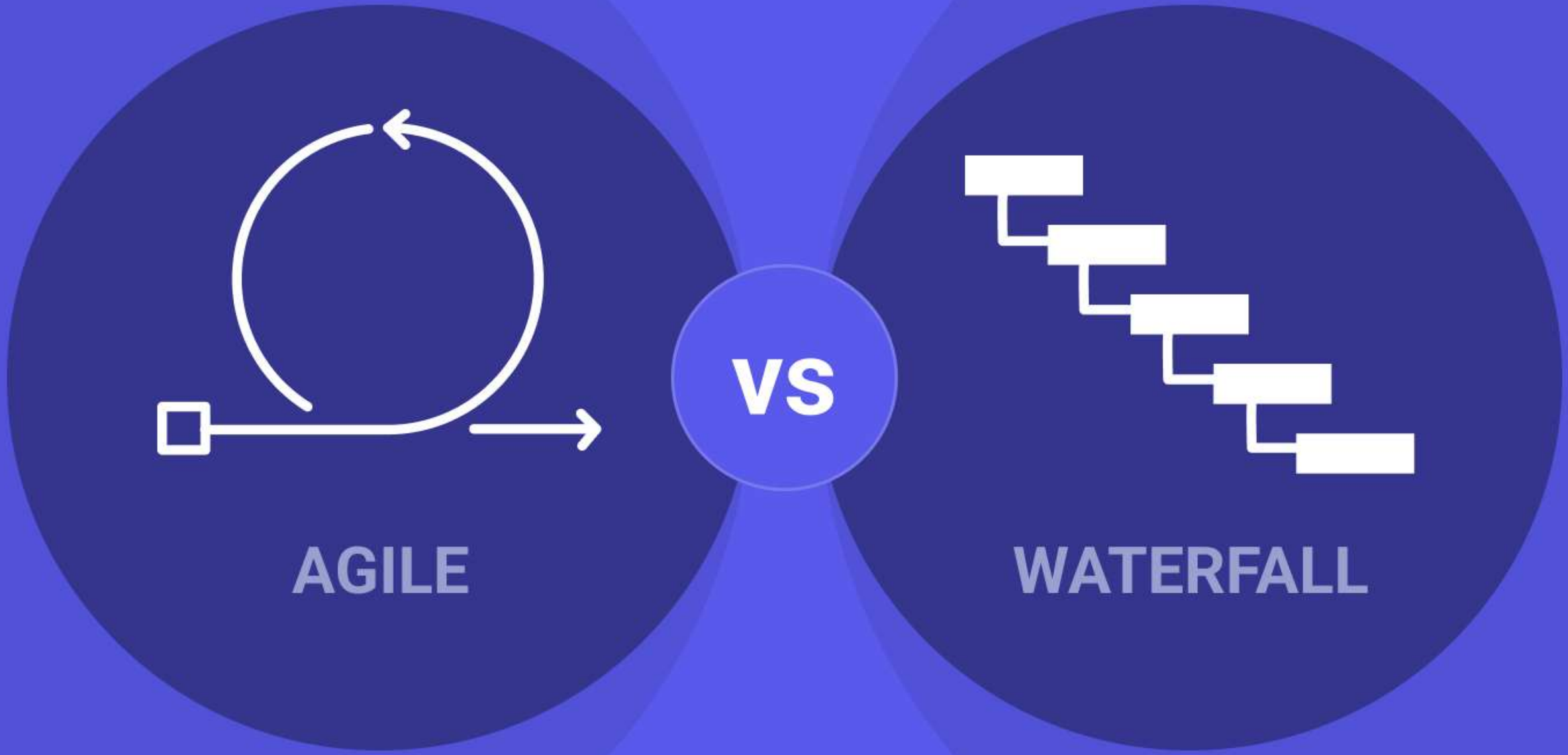
Source: Sandy Millar on Unsplash

Source: Muhammed ÖÇAL on Unsplash

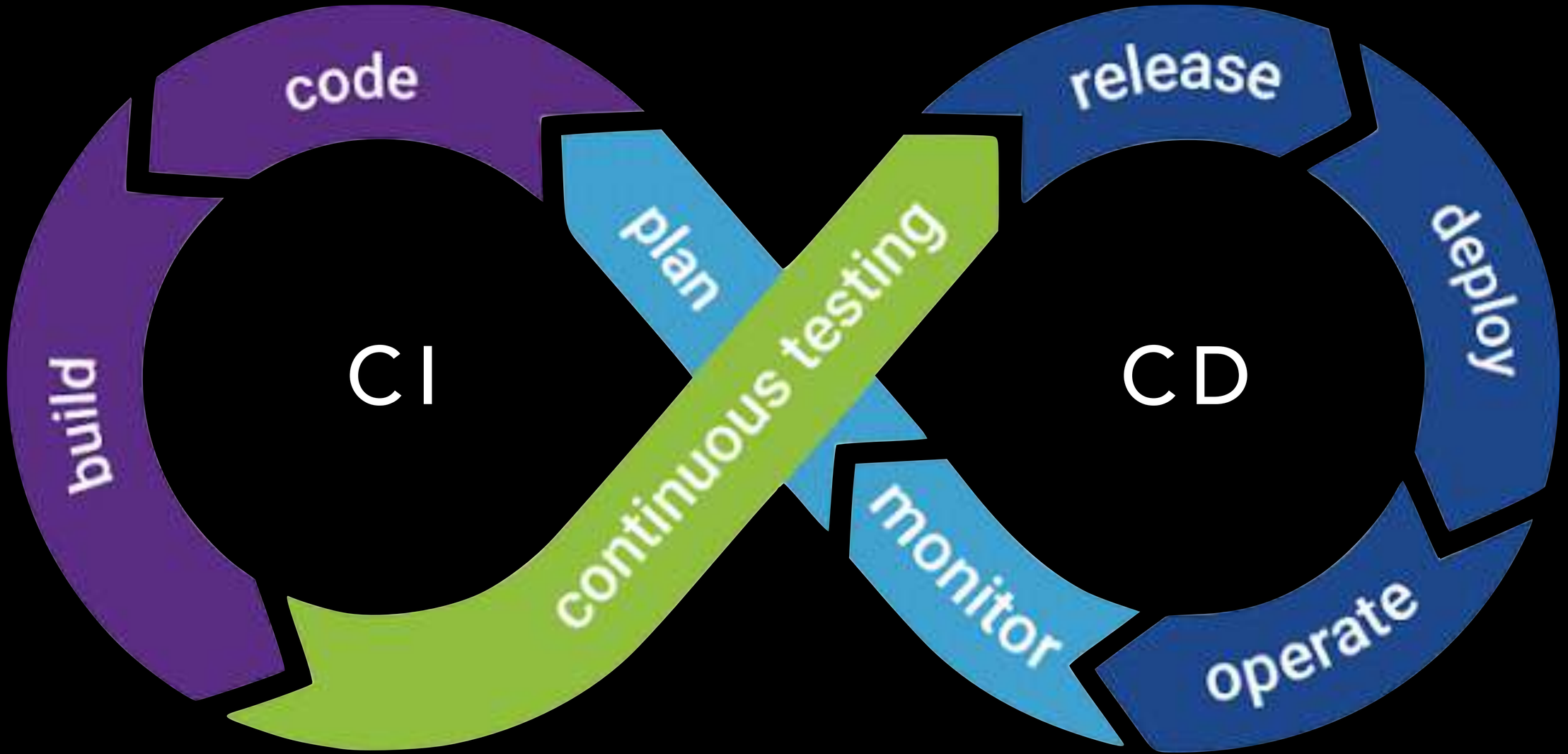


THERE'S LOTS THAT'S NOT SO NEW!

Source: Agile vs Waterfall: Differences, Pros & When to Use | Katalon



Source: What is CI/CD and How Does it Work? | Black Duck

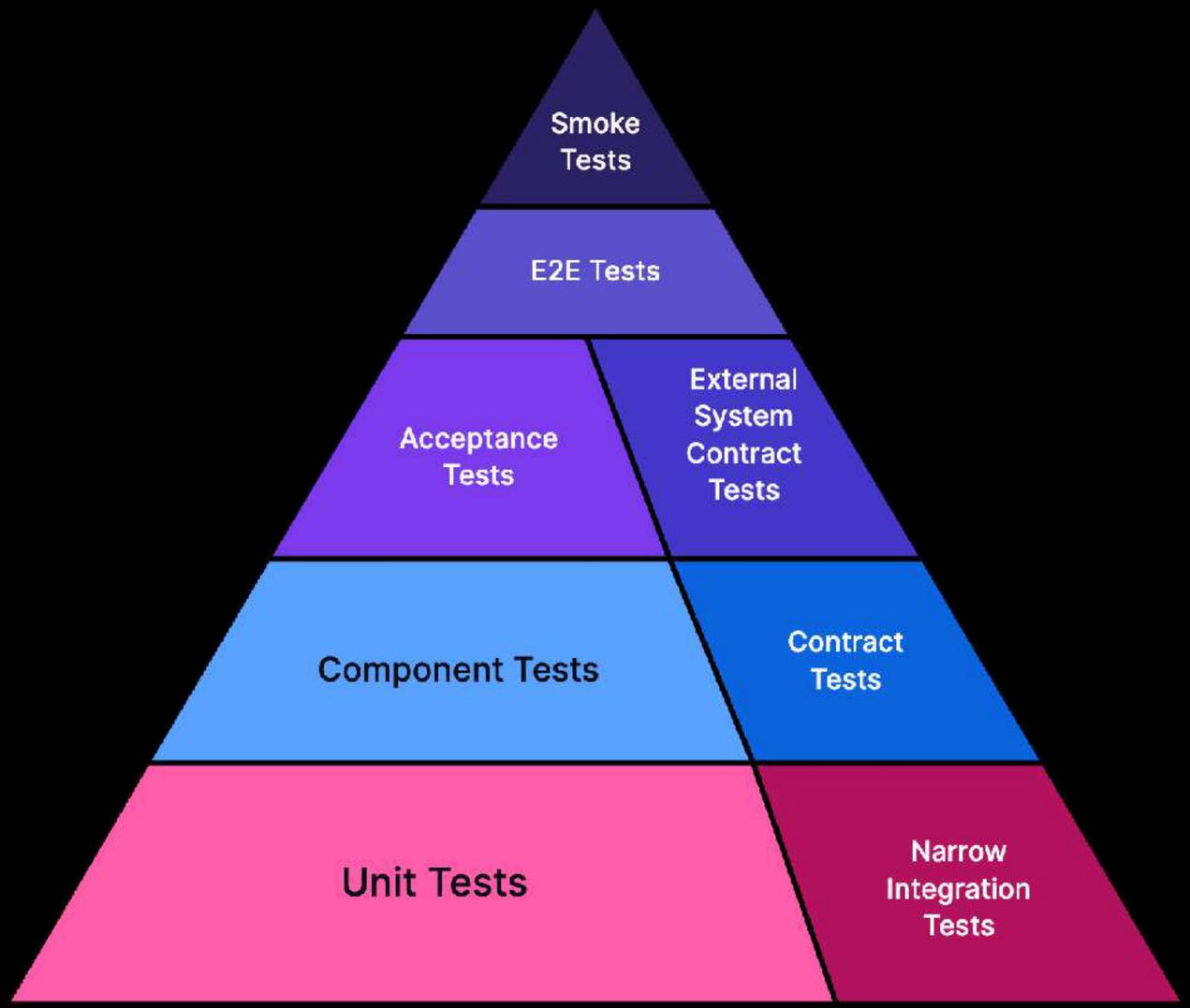




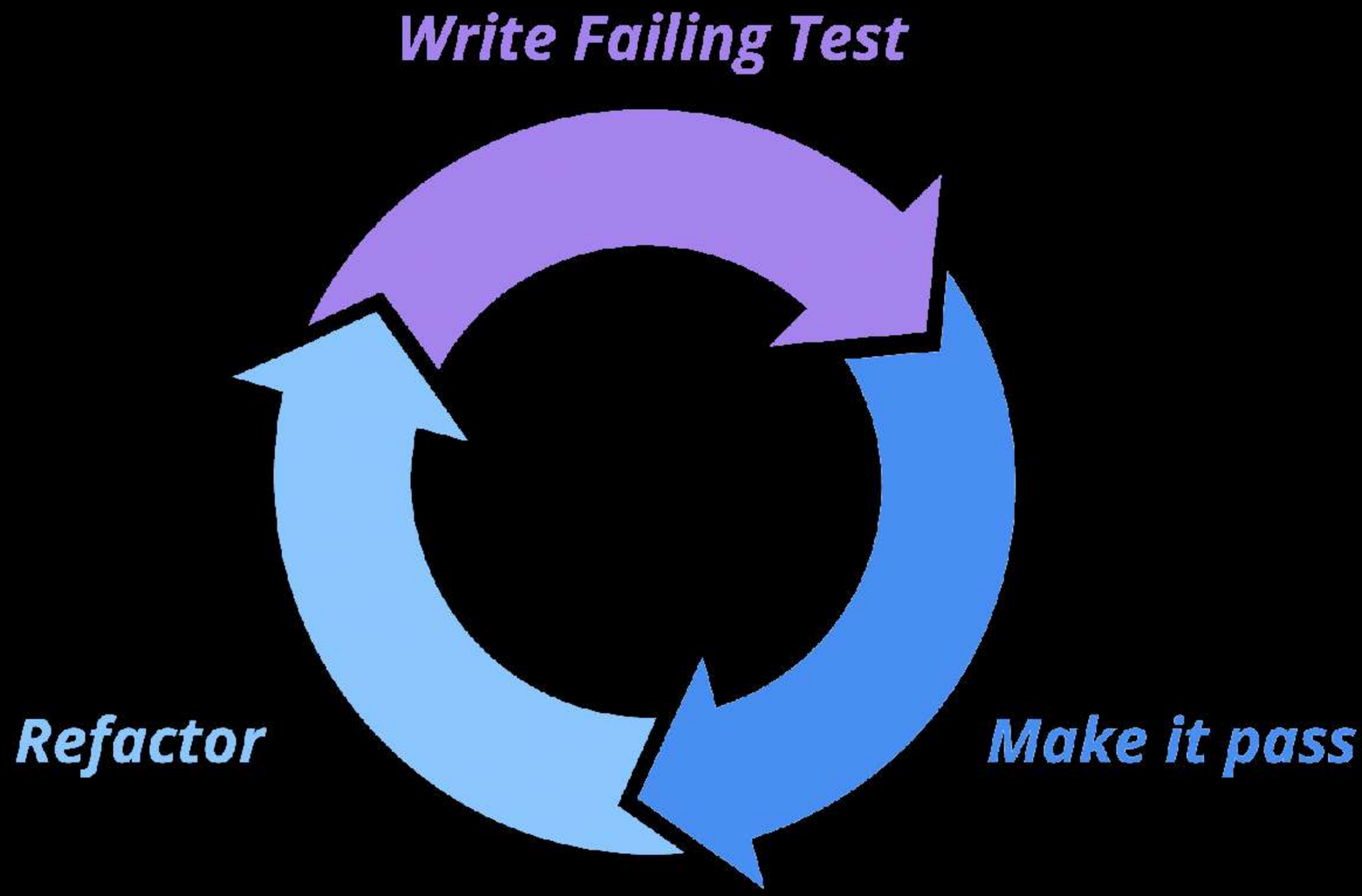
Source: CDC on Unsplash

Source: Modern Test Pyramid | Valentina (Cupać) Jemuović for Optivem

WE STILL NEED DIFFERENT TESTING TYPES

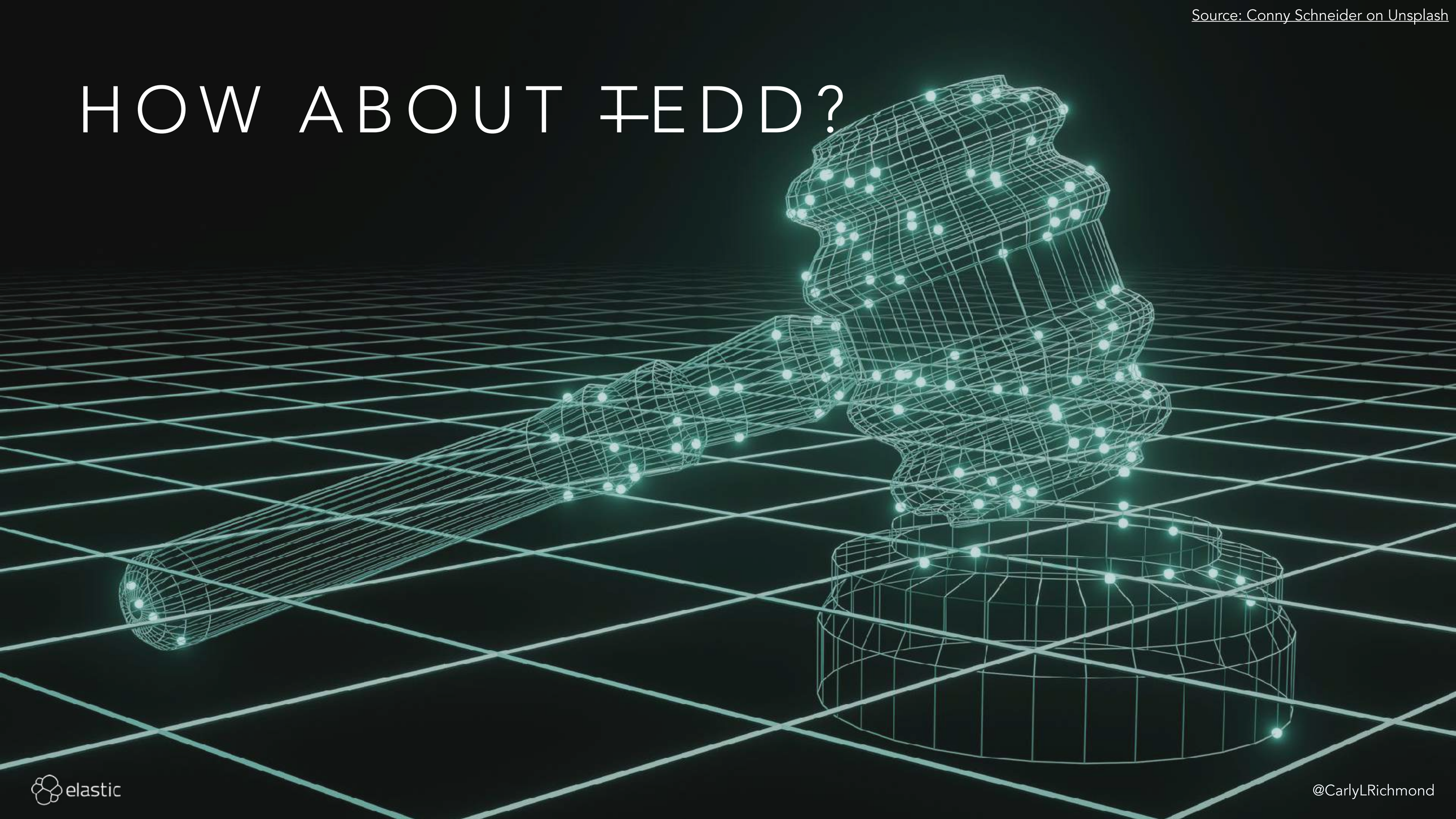


REMEMBER TDD?



Source: Conny Schneider on Unsplash

HOW ABOUT ~~T~~EDD?

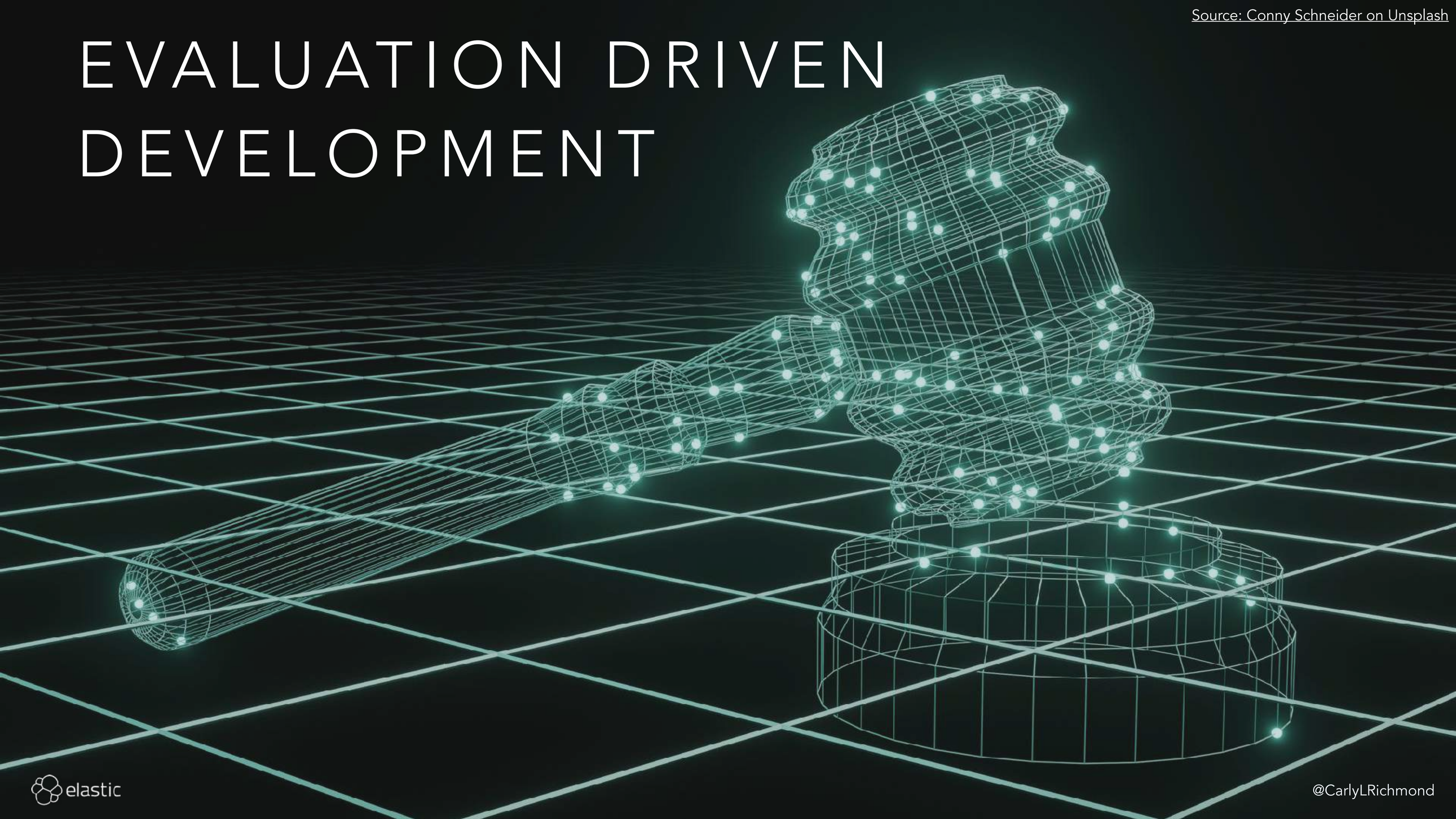


elastic

@CarlyLRichmond

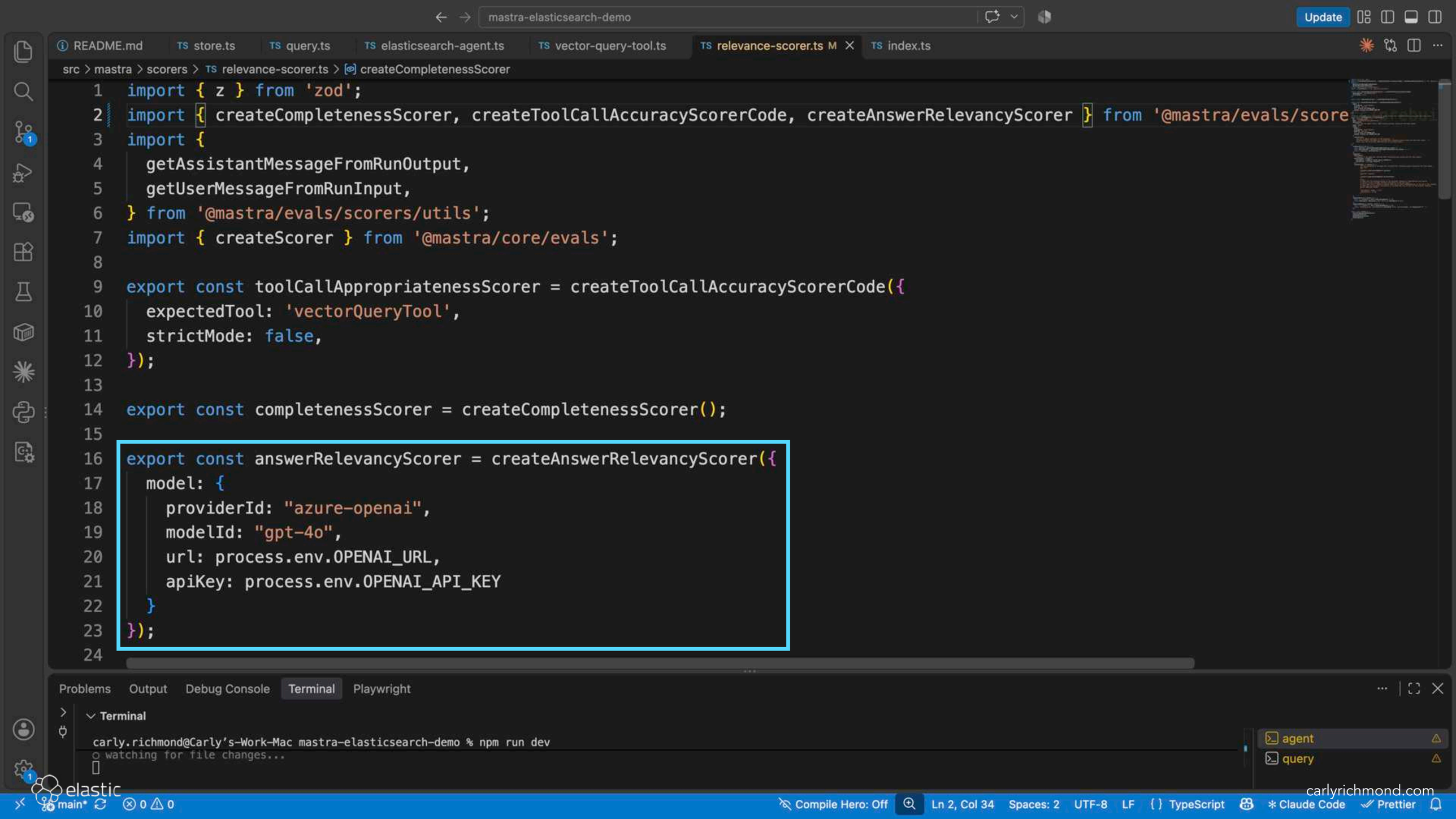
Source: Conny Schneider on Unsplash

EVALUATION DRIVEN DEVELOPMENT



elastic

@CarlyLRichmond


mastra-elasticsearch-demo
Update
README.md
store.ts
query.ts
elasticsearch-agent.ts
vector-query-tool.ts
relevance-scorer.ts M
index.ts
src > mastra > scorers > relevance-scorer.ts > createCompletenessScorer
import { z } from 'zod';
import { createCompletenessScorer, createToolCallAccuracyScorerCode, createAnswerRelevancyScorer } from '@mastra/evals/score
import {
getAssistantMessageFromRunOutput,
getUserMessageFromRunInput,
} from '@mastra/evals/scorers/utils';
import { createScorer } from '@mastra/core/evals';
export const toolCallAppropriatenessScorer = createToolCallAccuracyScorerCode({
expectedTool: 'vectorQueryTool',
strictMode: false,
});
export const completenessScorer = createCompletenessScorer();
export const answerRelevancyScorer = createAnswerRelevancyScorer({
model: {
providerId: "azure-openai",
modelId: "gpt-4o",
url: process.env.OPENAI_URL,
apiKey: process.env.OPENAI_API_KEY
}
});
Problems
Output
Debug Console
Terminal
Playwright
Terminal
carly.richmond@Carly's-Work-Mac mastra-elasticsearch-demo % npm run dev
watching for file changes...
agent
query
elastic
carlyrichmond.com
main*
0 0
Compile Hero: Off
Ln 2, Col 34
Spaces: 2
UTF-8
LF
TypeScript
Claude Code
Prettier

Search ⌘ K

Primitives
Agents
Workflows
Processors
MCP Servers
Tools
Workspaces
Request Context

Evaluation
Overview
Scorers
Settings
Resources

All Entities

Date	Time	Input
Jul 21	3:04:46 pm	{"inputMessages":[{"id":"f9c819be
Jul 01	5:34:03 pm	{"inputMessages":[{"id":"1cb2c55e
Jul 01	5:33:42 pm	{"inputMessages":[{"id":"c8bc9149
Jul 01	5:32:52 pm	{"inputMessages":[{"id":"3025eb50
Jul 01	5:32:24 pm	{"inputMessages":[{"id":"c4785ba0
Jul 01	5:31:33 pm	{"inputMessages":[{"id":"7a7c5fb8
Jul 01	5:27:00 pm	{"inputMessages":[{"id":"29770eef
Jul 01	5:26:06 pm	{"inputMessages":[{"id":"82834de3
Jul 01	3:45:55 pm	{"inputMessages":[{"id":"fda82482
Jul 01	3:40:03 pm	{"inputMessages":[{"id":"96e1228b
Jul 01	2:11:44 pm	{"inputMessages":[{"id":"30e6da0d
Jul 01	1:44:47 pm	{"inputMessages":[{"id":"7becc3a6
Jul 01	1:43:28 pm	{"inputMessages":[{"id":"b3f16650
Jul 01	1:40:54 pm	{"inputMessages":[{"id":"2b64806b
Jul 01	1:40:04 pm	{"inputMessages":[{"id":"0a450337

Score # e810864b-1eac-42d2-bacf-5859da40bad0

Scorer Answer Relevancy Scorer
Created Jul 01, 17:33:42.659
Trace Id b9156d34b5a533f286d579b112f5c00a
Span Id c8e2a6ab99eda0e2

Save as Dataset Item

Score: **0.54**

The score is 0.54 because the output includes specific examples of UFO-related movies, which directly address part of the input. However, the methodology and descriptions add contextual relevance without fully responding to a specific query. Mixed aspects, such as mentioning a TV series and outlining thought processes, offer partial relevance but dilute the direct pertinence to UFO movies. Overall, the balance of direct responses and tangential additions results in a moderate score.

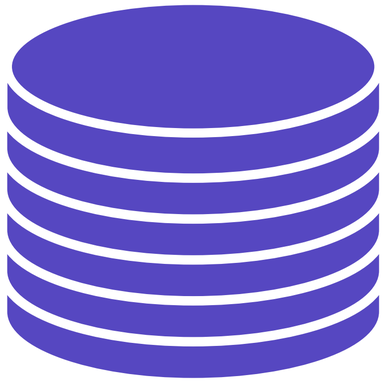
Source: mostafa meraji on Unsplash

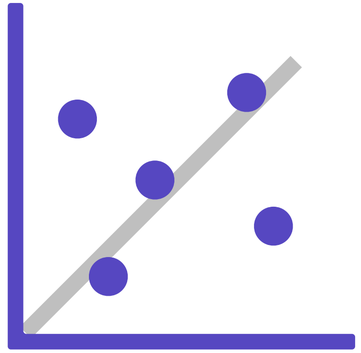


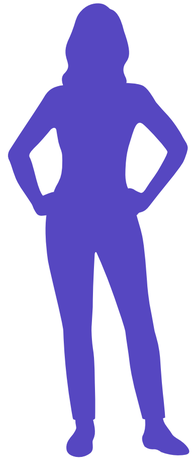
DATA DISCOVERY

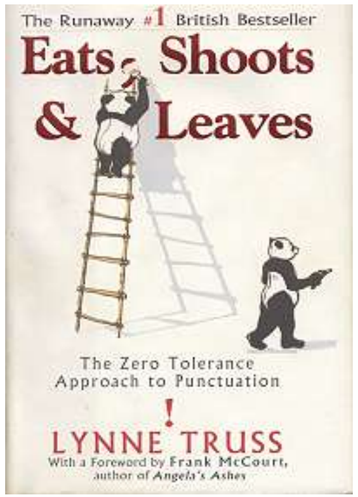
“Rubbish in, rubbish out!”

–CARLY’S HIGH SCHOOL MATHS TEACHER








The Runaway #1 British Bestseller
Eats, Shoots & Leaves
The Zero Tolerance Approach to Punctuation
LYNNE TRUSS
With a Foreword by Frank McCourt, author of Angela's Ashes



Source: OpenAI | Why Language Models Hallucinate

Why Language Models Hallucinate

Adam Tauman Kalai
OpenAI

Ofir Nachum
OpenAI

Santosh S. Vempala
Georgia Tech

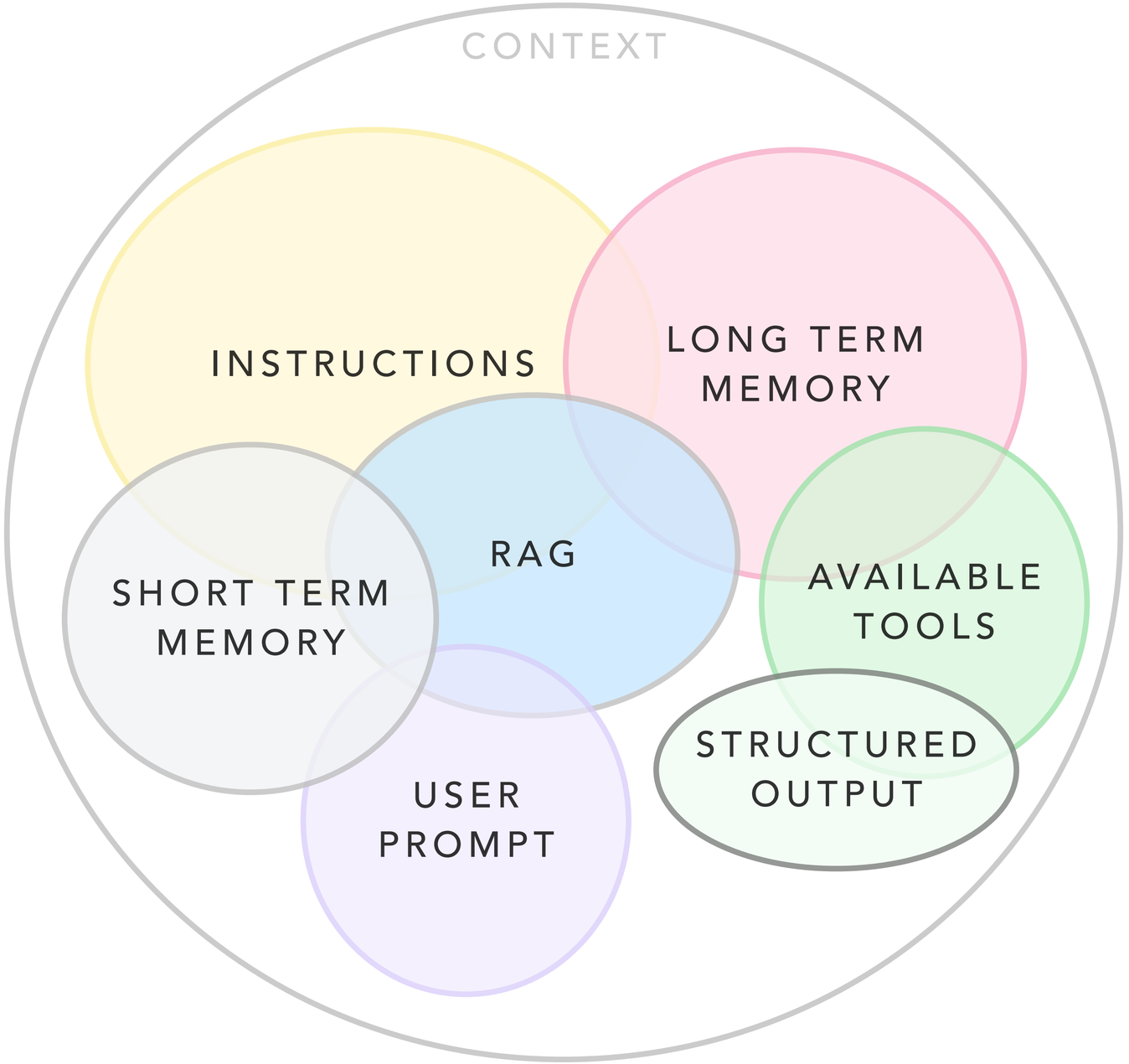
Edwin Zhang
OpenAI

September 4, 2025

Abstract

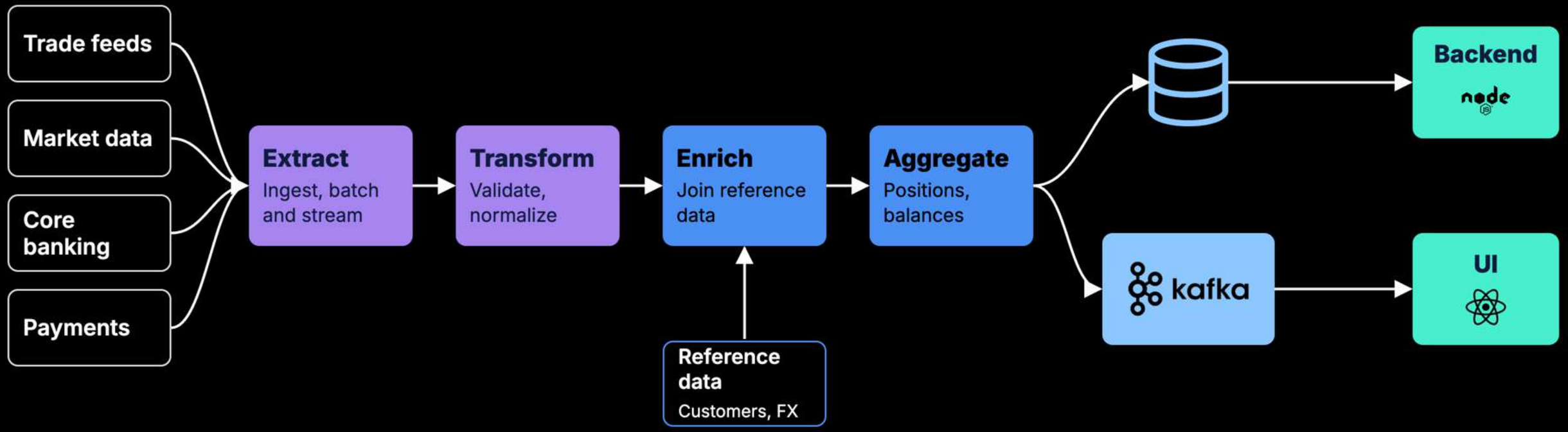
Like students facing hard exam questions, large language models sometimes guess when uncertain, producing plausible yet incorrect statements instead of admitting uncertainty. Such "hallucinations" persist even in state-of-the-art systems and undermine trust. We argue that language models hallucinate because the training and evaluation procedures reward guessing over acknowledging uncertainty, and we analyze the statistical causes of hallucinations in the modern training pipeline. Hallucinations need not be mysterious—they originate simply as errors in binary classification. If incorrect statements cannot be distinguished from facts, then hallucinations in pretrained language models will arise through natural statistical pressures. We then argue that hallucinations persist due to the way most evaluations are graded—language models are optimized to be good test-takers, and guessing when uncertain improves test performance. This "epidemic" of penalizing uncertain responses can only be addressed through a socio-technical mitigation: modifying the scoring of existing benchmarks that are misaligned but dominate leaderboards, rather than introducing additional hallucination evaluations. This change may steer the field toward more trustworthy AI systems.

1 Introduction



https://www.elastic.co/search-labs/blog/context-engineering-overview

WE'VE ALWAYS HAD TO FIND THE RIGHT DATA




Park Tool
TOBE
BIKE HAND
BIKE HAND
Park Tool
Park Tool
Park Tool

REMEMBER  OPENAPI ?

INITIATIVE

Close a point in time DELETE
Count search results GET
Explain a document match result POST
Get the field capabilities POST
Run multiple searches POST
Run multiple templated searches POST
Open a point in time POST
Evaluate ranked search results POST
Render a search template POST
Run a search POST
Search a vector tile GET
Get the search shards POST
Run a search with a search template POST
Get terms in an index POST
Search application
Searchable snapshots
Security
Snapshot and restore
Snapshot lifecycle management
SQL

Run a search

Ask AI | Generally available

`POST /{index}/_search`

Api key auth | Basic auth | Bearer auth

All methods and paths for this operation:

- `GET /_search`
- `POST /_search`
- `GET /{index}/_search`
- `POST /{index}/_search`

Get search hits that match the query defined in the request. You can provide search queries using the `q` query string parameter or the request body. If both are specified, only the query parameter is used.

If the Elasticsearch security features are enabled, you must have the read index privilege for the target data stream, index, or alias. For cross-cluster search, refer to the documentation about configuring CCS privileges. To search a point in time (PIT) for an alias, you must have the `read` index privilege for the alias's data streams or indices.

Search slicing

POST /{index}/_search — Console

```
GET /my-index-000001/_search?from=40&size=20
{
  "query": {
    "term": {
      "user.id": "kimchy"
    }
  }
}
```

Request examples — A simple term search

```
// Run `GET /my-index-000001/_search?from=40&size=20` to run a
// search.
{
  "query": {
    "term": {
      "user.id": "kimchy"
    }
  }
}
```

Response examples (200)

carlyrichmond.com

weather.tool.ts

```ts
import { tool } from 'ai';
import { z } from 'zod';

import { WeatherResponse } from '../model/weather.model';

export const weatherTool = tool({
  description:
  'Display the weather for a holiday location',
  inputSchema: z.object({
    location: z.string().describe('The location to get the weather for')
  }),
  outputSchema: z.object({
    location: z.string().describe('The location to get the weather for'),
    condition: z.string().optional().describe('The current weather condition'),
    condition_image: z.string().optional().describe('An image representing the current weather condition'),
    temperature: z.number().optional().describe('The current temperature in Celsius'),
    feels_like_temperature: z.number().optional().describe('What the temperature feels like in Celsius'),
    humidity: z.number().optional().describe('The current humidity percentage'),
    message: z.string().optional().describe('An optional message, e.g. for errors')
  }),
  execute: async ({ location }: { location: string}) => {
    // While a historical forecast may be better, this example gets the next 3 days
    const url = `https://api.weatherapi.com/v1/forecast.json?q=${location}&days=3&key=${process.env.WEATHER_API_KEY}`;

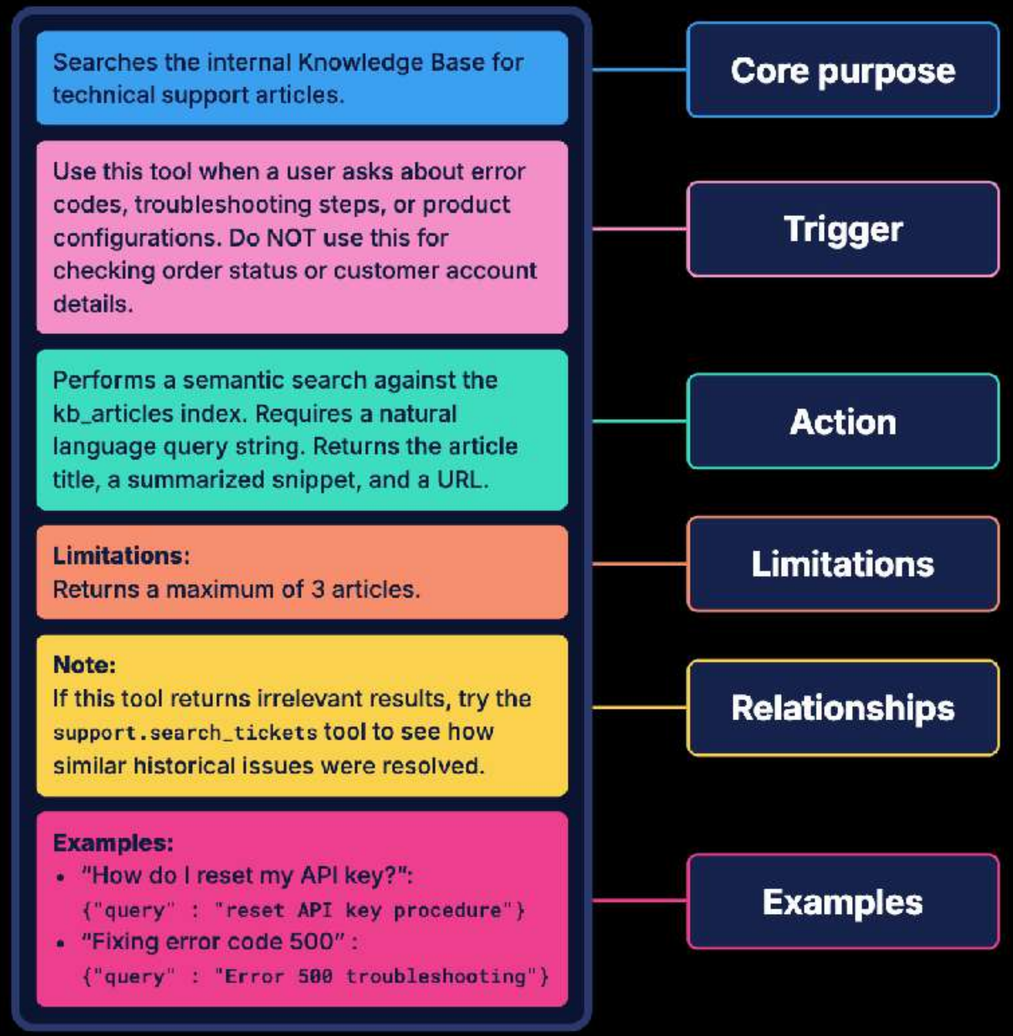
    try {
      const response = await fetch(url);
      const weather : WeatherResponse = await response.json();
      return {
        location: location,
        condition: weather.current.condition.text,
        condition_image: weather.current.condition.icon,
        temperature: Math.round(weather.current.temp_c),
        feels_like_temperature: Math.round(weather.current.feelslike_c),
        humidity: weather.current.humidity
      };
    } catch(e) {
      console.error(e);
      return {
        message: 'Unable to obtain weather information',
        location: location
      };
    }
  }
});
```

Credit: Leonie Monigatti | AI Engineer Agentic Search Workshop

HOW TO MAKE AGENTS CALL THE CORRECT TOOL

1. Prompt engineer the **tool description**

2. Reinforce in the agent's **system prompt**



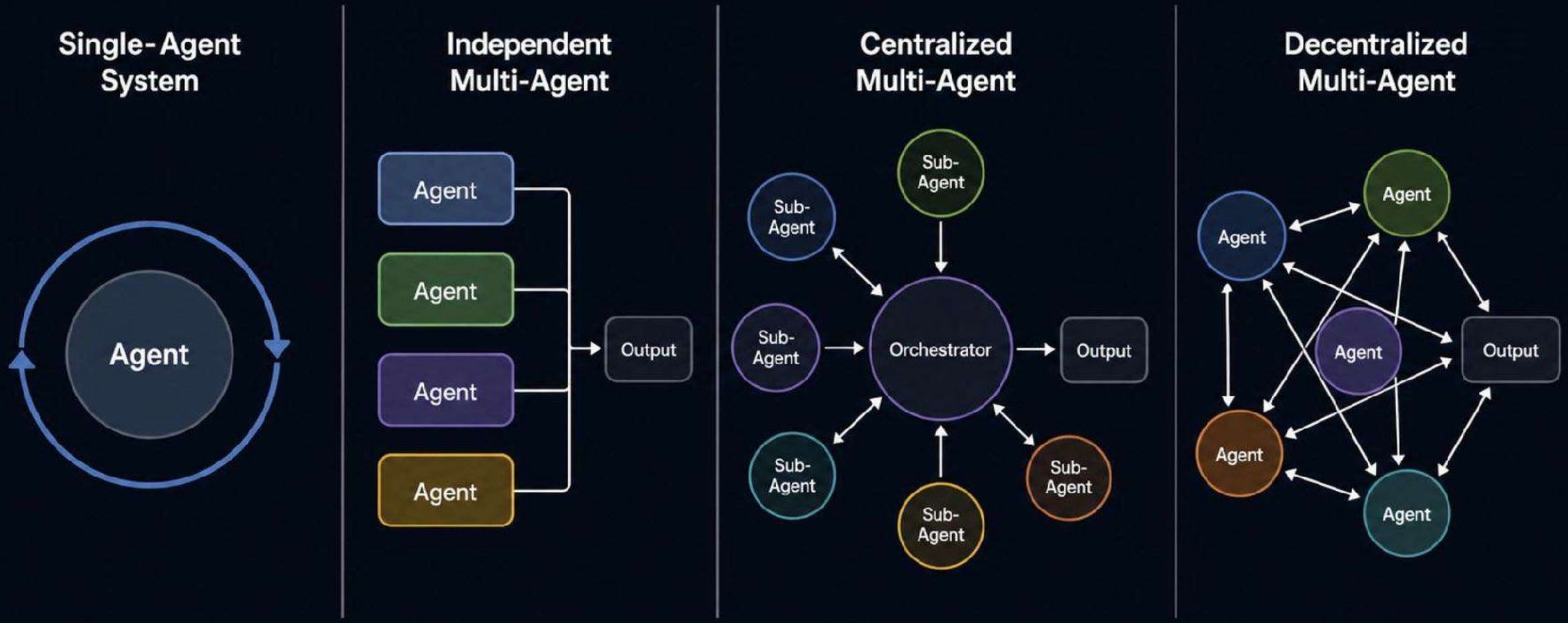
ORCHESTRATION



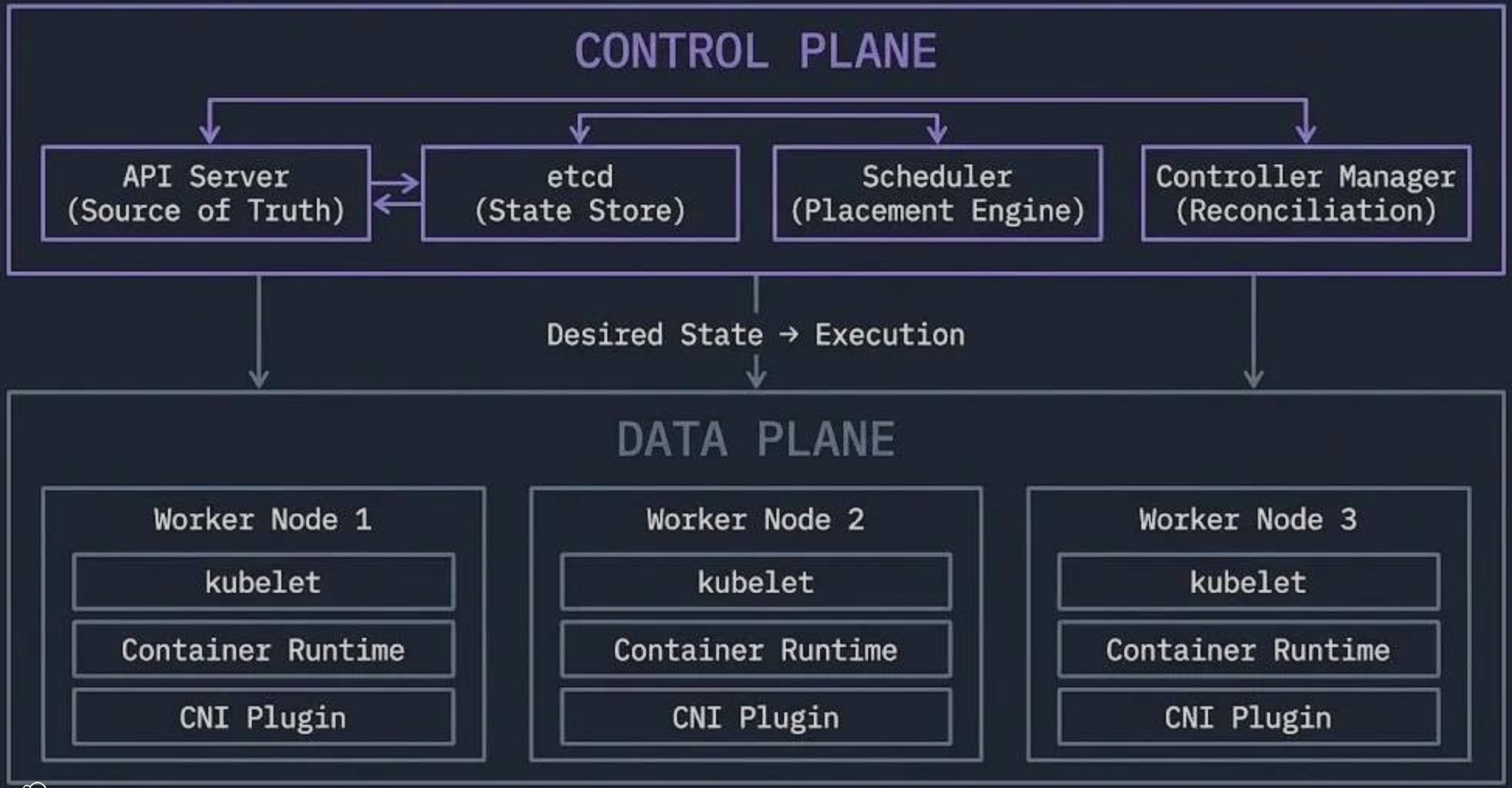
Source: Robert Katzki on Unsplash

Source: Four Agent Orchestration Patterns You Should Know About | Jim Clyde Monge @ AlphaSignal

AGENT ORCHESTRATION



Source: Kubernetes Cloud Orchestration | RACK2CLOUD





Source: Andres Siimon on Unsplash

MONITORING & OBSERVABILITY

records-ui-web

Alerts SLOs Add data

Overview Transactions Dependencies Errors Metrics Infrastructure Logs Alerts Dashboards



Trace sample

2 of 500

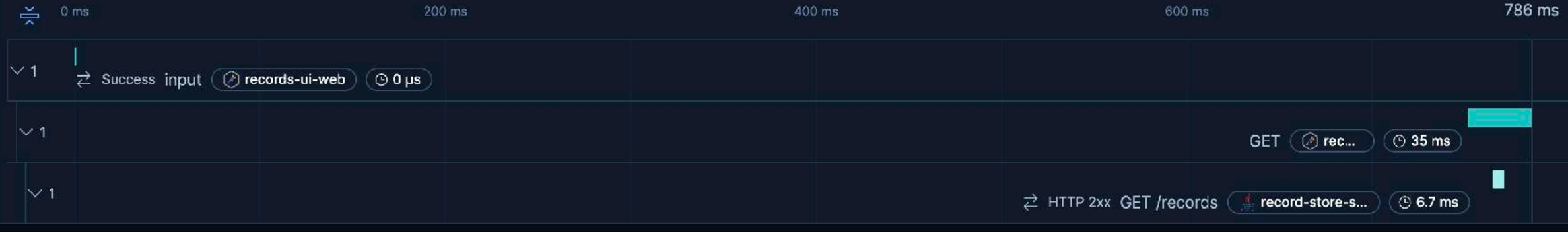
View full trace | Open full trace in Discover | Investigate

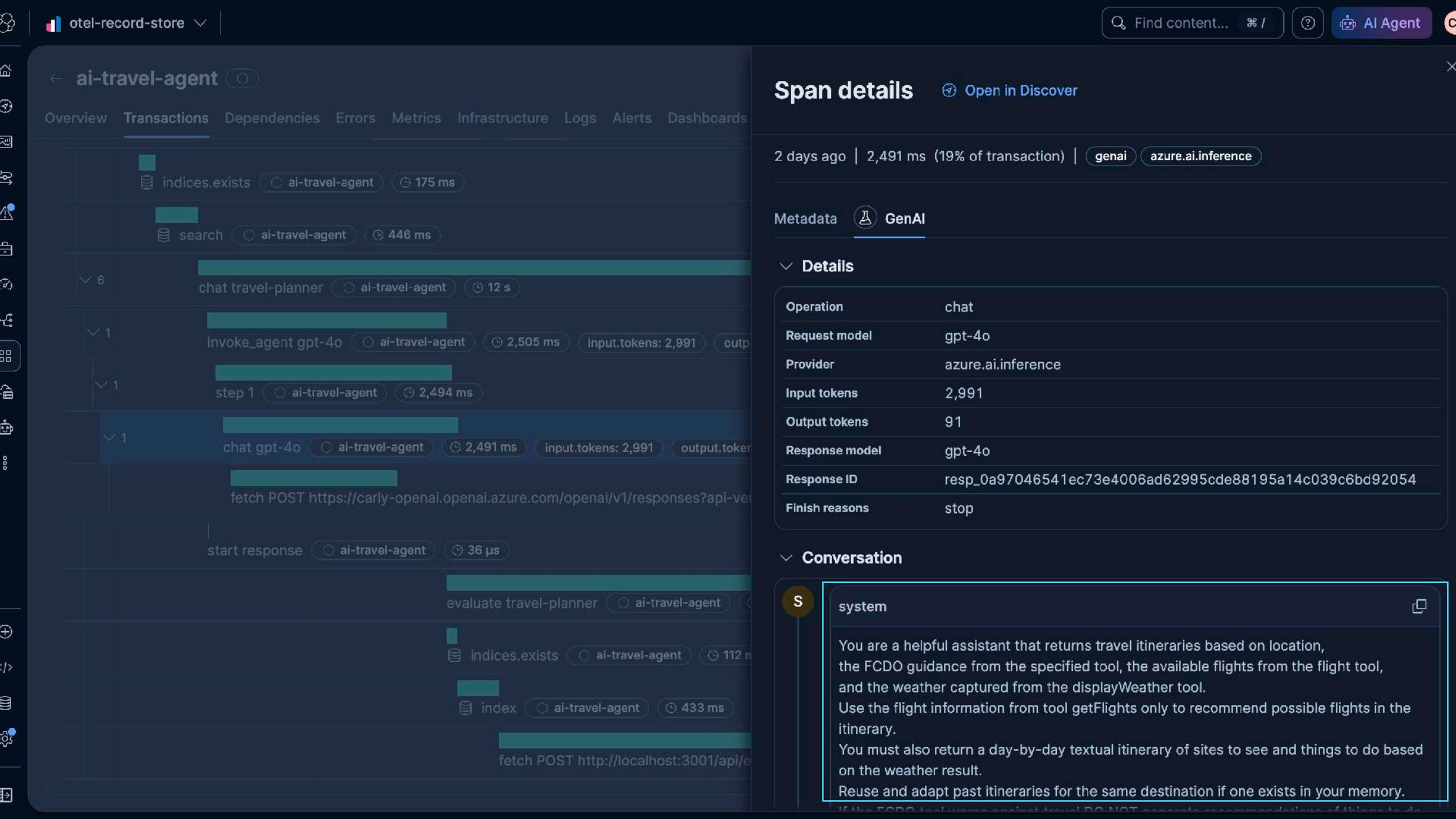
7 days ago | 0 µs 0 | http://localhost:4173/records

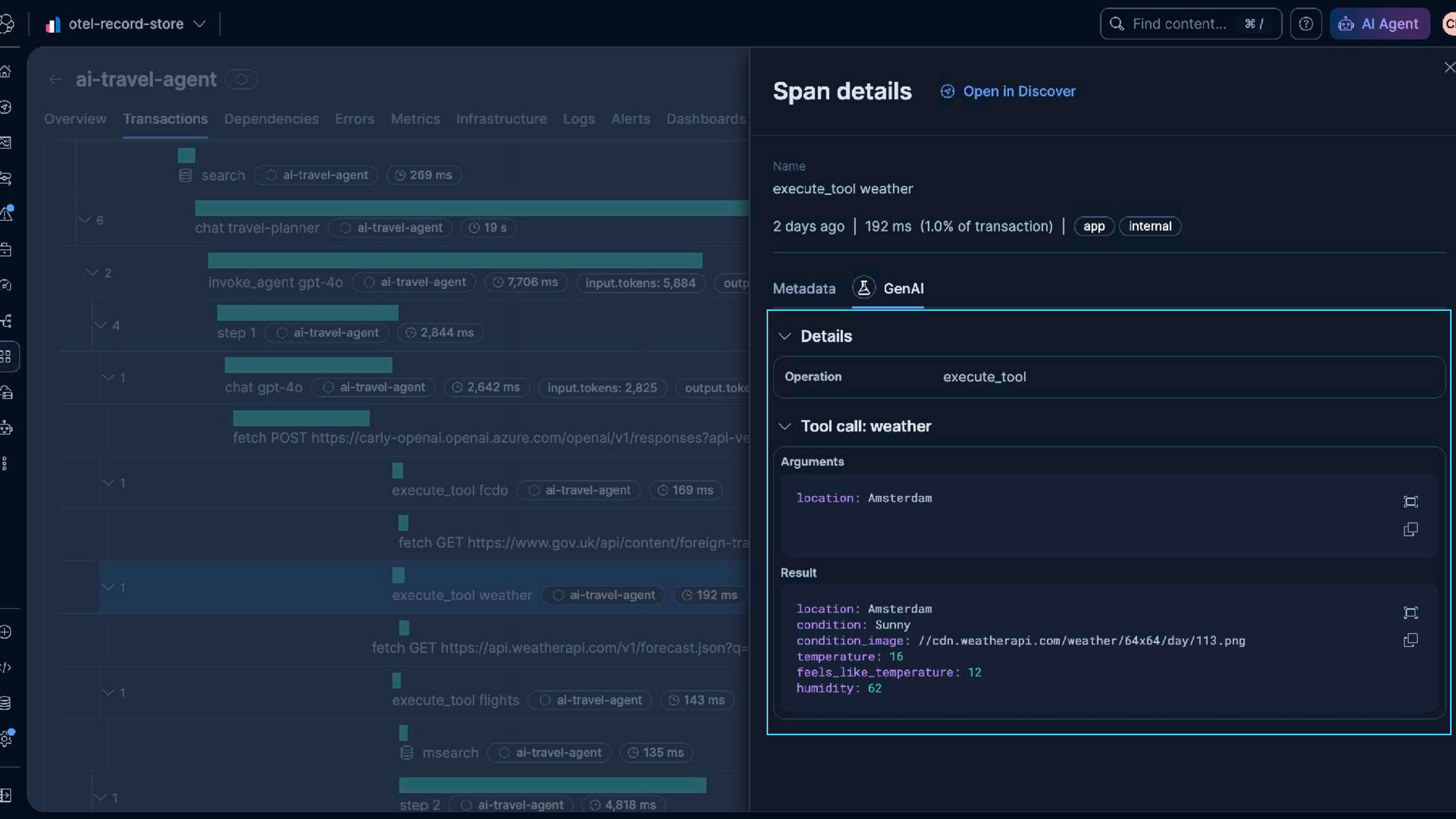
Timeline Metadata Logs

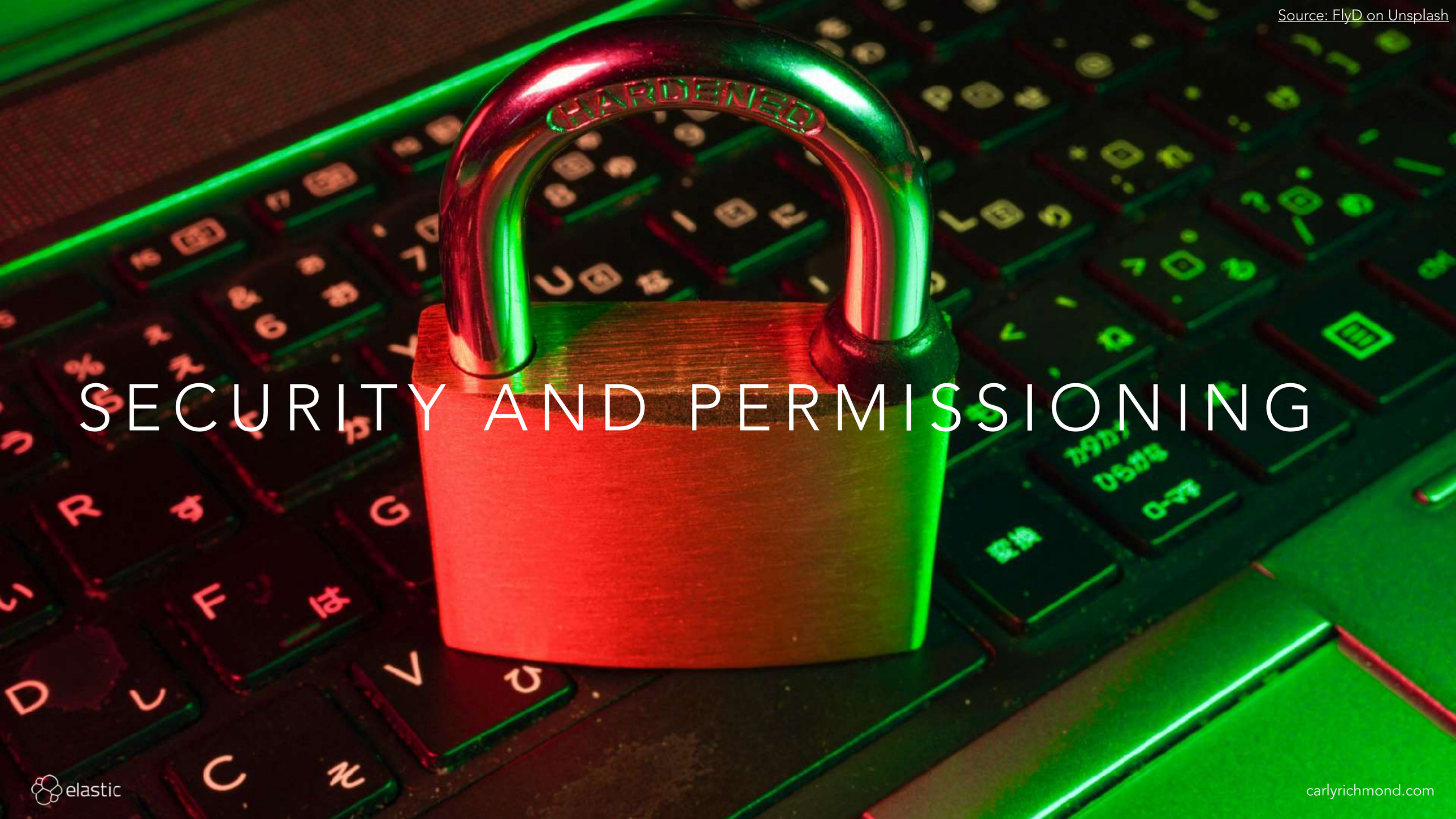
Show critical path

Services records-ui-web record-store-server-java




otel-record-store
Find content... ⌘ /
AI Agent
ai-travel-agent
Overview
Transactions
Dependencies
Errors
Metrics
Infrastructure
Logs
Alerts
Dashboards
indices.exists ai-travel-agent 175 ms
search ai-travel-agent 446 ms
6
chat travel-planner ai-travel-agent 12 s
1
invoke_agent gpt-4o ai-travel-agent 2,505 ms input.tokens: 2,991
1
step 1 ai-travel-agent 2,494 ms
1
chat gpt-4o ai-travel-agent 2,491 ms input.tokens: 2,991
fetch POST https://carly-openai.openai.azure.com/openai/v1/responses?api-ver
start response ai-travel-agent 36 µs
evaluate travel-planner ai-travel-agent
indices.exists ai-travel-agent
index ai-travel-agent 433 ms
fetch POST http://localhost:3001/api/e
Span details
Open in Discover
2 days ago | 2,491 ms (19% of transaction) | genai azure.ai.inference
Metadata
GenAI
Details
Operation chat
Request model gpt-4o
Provider azure.ai.inference
Input tokens 2,991
Output tokens 91
Response model gpt-4o
Response ID resp_0a97046541ec73e4006ad62995cde88195a14c039c6bd92054
Finish reasons stop
Conversation
S
system
You are a helpful assistant that returns travel itineraries based on location,
the FCDO guidance from the specified tool, the available flights from the flight tool,
and the weather captured from the displayWeather tool.
Use the flight information from tool getFlights only to recommend possible flights in the itinerary.
You must also return a day-by-day textual itinerary of sites to see and things to do based on the weather result.
Reuse and adapt past itineraries for the same destination if one exists in your memory.


otel-record-store
Find content... ⌘ /
AI Agent
ai-travel-agent
Overview
Transactions
Dependencies
Errors
Metrics
Infrastructure
Logs
Alerts
Dashboards
search ai-travel-agent 269 ms
6
chat travel-planner ai-travel-agent 19 s
2
invoke_agent gpt-4o ai-travel-agent 7,706 ms input.tokens: 5,884
4
step 1 ai-travel-agent 2,844 ms
1
chat gpt-4o ai-travel-agent 2,642 ms input.tokens: 2,825
fetch POST https://carly-openai.openai.azure.com/openai/v1/responses?api-ve
1
execute_tool fcdo ai-travel-agent 169 ms
fetch GET https://www.gov.uk/api/content/foreign-tra
1
execute_tool weather ai-travel-agent 192 ms
fetch GET https://api.weatherapi.com/v1/forecast.json?q=
1
execute_tool flights ai-travel-agent 143 ms
msearch ai-travel-agent 135 ms
1
step 2 ai-travel-agent 4,818 ms
Span details
Open in Discover
Name
execute_tool weather
2 days ago | 192 ms (1.0% of transaction) | app internal
Metadata
GenAI
Details
Operation execute_tool
Tool call: weather
Arguments
location: Amsterdam
Result
location: Amsterdam
condition: Sunny
condition_image: //cdn.weatherapi.com/weather/64x64/day/113.png
temperature: 16
feels_like_temperature: 12
humidity: 62



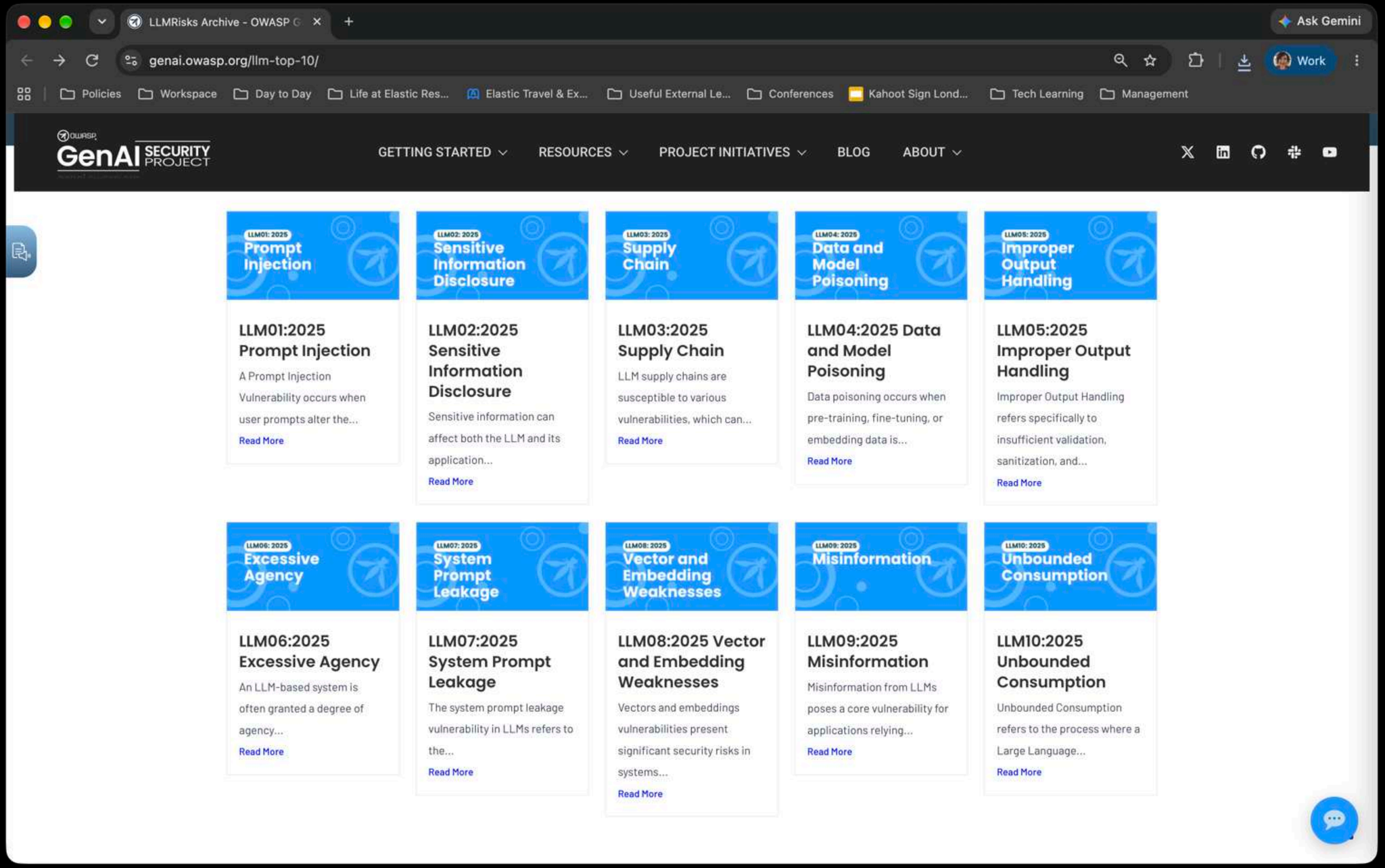
Source: FlyD on Unsplash

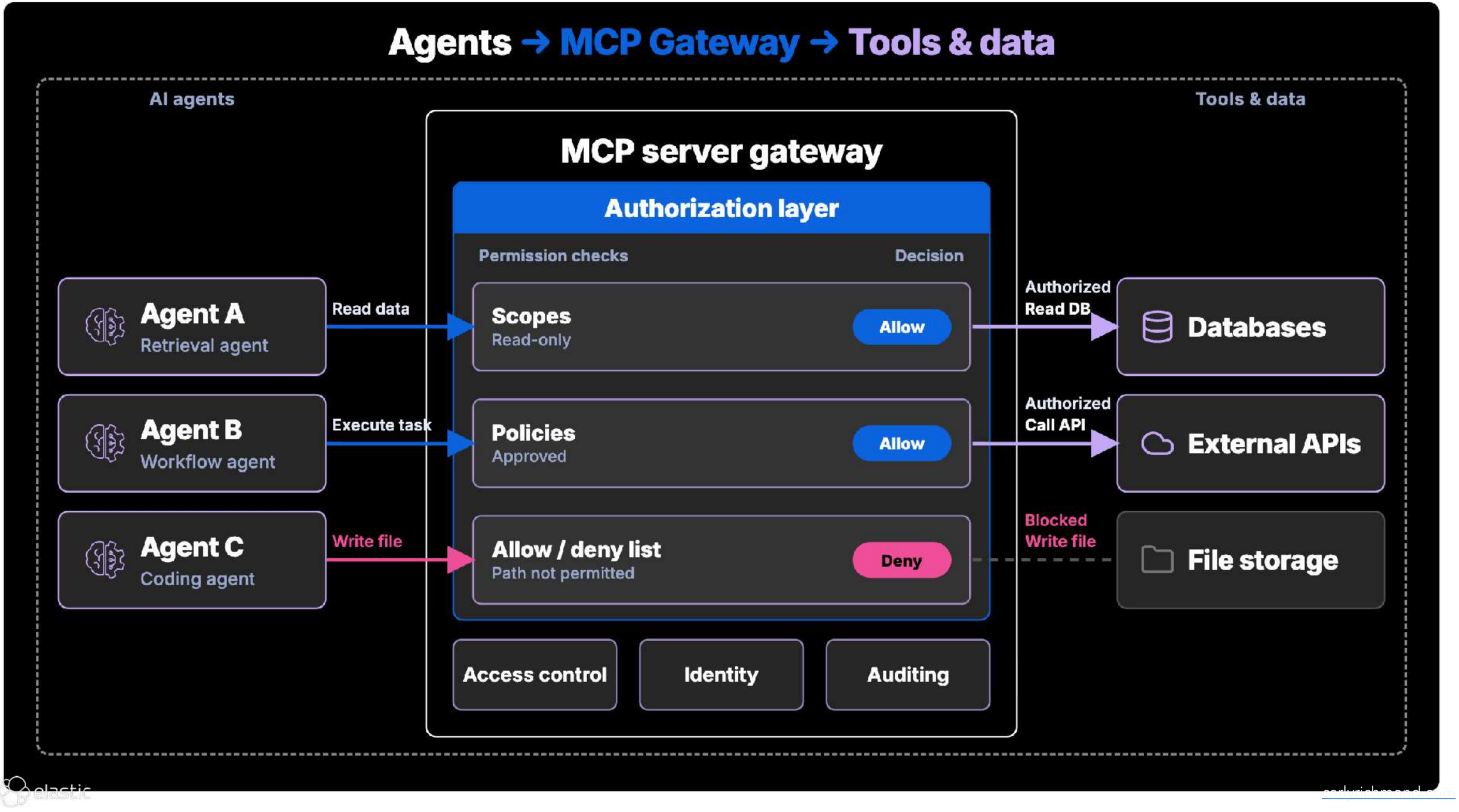
SECURITY AND PERMISSIONING

Source: What Are OWASP Top 10 Threads & When Will the Top 10 Be Updated? | Tiffany Jennings on Mend.io



Source: OWASP Top 10 for Large Language Model Applications




Agents → MCP Gateway → Tools & data
AI agents
Tools & data
MCP server gateway
Authorization layer
Permission checks
Decision
Agent A
Retrieval agent
Read data
Scopes
Read-only
Allow
Authorized
Read DB
Databases
Agent B
Workflow agent
Execute task
Policies
Approved
Allow
Authorized
Call API
External APIs
Agent C
Coding agent
Write file
Allow / deny list
Path not permitted
Deny
Blocked
Write file
File storage
Access control
Identity
Auditing

otel-record-store

Find content... ⌘ /

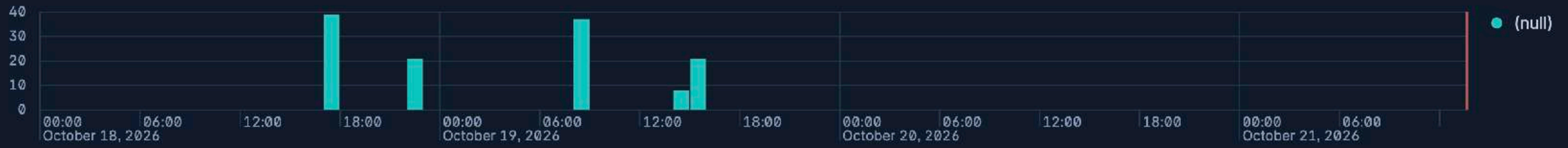
AI Agent

New session | Evals | Guardrails

Query in ES|QL | Save

Data view | All logs (OpenLit Demo Oct 2026) | attributes.guard.score >= 0 | Last 3 days (rounded) | 00:32 | Refresh

Auto interval | Breakdown by lo...evel



Oct 18, 2026 @ 00:00:00.000 - Oct 21, 2026 @ 13:38:57.790 (interval: Auto - hour)

126 documents | View as

Columns 7 | Sort fields 1

Actions	@timestamp	attributes.guard.name	attributes.guard.phase	attributes.guard.action	attributes.guard.explanation	attributes.guard.score	attributes.guard.latency_ms
	Oct 19, 2026 @ 15:36:28.690	topic_restriction	preflight	deny	Topic 'politics' is not in the allowed list	1	2.8
	Oct 19, 2026 @ 15:36:28.687	pii	preflight	allow	(blank)	0	0.1
	Oct 19, 2026 @ 15:36:28.687	prompt_injection	preflight	allow	(blank)	0	0.0
	Oct 19, 2026 @ 15:36:28.686	sensitive_topic	preflight	allow	(blank)	0	0.0
	Oct 19, 2026 @ 15:36:28.686	moderation	preflight	allow	(blank)	0	0.0

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Source: JD-Photos on Unsplash



Source: Sandy Millar on Unsplash

ABOUT LEARNING EVERYTHING!

Source: Wilhelm Gunkel on Unsplash

